

cap set problem

Vincent Wang

July 2026

1 Upper Bounds

Let $\mathbb{F}_q$ be a finite field and $A \subseteq \mathbb{F}_q^n$. Let M_n denote the set of monomials in n variables with each exponent less than q and P_n the span of M_n . For real $d \in [0, n(q-1)]$, let M_n^d denote the set of monomials in M_n with degree at most d and let P_n^d denote the span. Let $m_d = |M_n^d|$.

Claim: Suppose that $\alpha, \beta, \gamma \in \mathbb{F}_q$ with sum zero and $P \in P_n^d$ such that $P(\alpha a + \beta b) = 0$ for all distinct $a, b \in A$. Then $P(-\gamma c) \neq 0$ for at most $2m_{d/2}$ values of c .

Proof: Define the $A \times A$ matrix B by

$$\begin{aligned} B_{a,b} &= P(\alpha a + \beta b) = \sum_{\substack{m, m' \in M_n^d \\ \deg mm' \leq d}} c_{m, m'} m(a) m'(b) \\ &= \sum_{m \in M_n^{d/2}} m(a) F_m(b) + \sum_{m \in M_n^{d/2}} G_m(a) m(b) \end{aligned}$$

Let $v_m = (m(a))_{a \in A}$, $f_m = (F_m(a))_{a \in A}$, $g_m = (G_m(a))_{a \in A}$. Then

$$B = \sum_{m \in M_n^{d/2}} v_m f_m^T + \sum_{m \in M_n^{d/2}} g_m v_m^T,$$

so B is the sum of at most $2m_{d/2}$ rank 1 matrices and thus at most rank $2m_{d/2}$.

By construction, B is diagonal. The diagonal entries of B are $B_{c,c} = P(\alpha c + \beta c) = P(-\gamma c)$. Since B is diagonal, there are at most $2m_{d/2}$ values for which $P(-\gamma c) \neq 0$.

Claim: Let $\alpha, \beta, \gamma \in \mathbb{F}_q$ not all zero with sum zero. Let A have the property $\alpha a + \beta b + \gamma c = 0$ for $a, b, c \in A$ only when $a = b = c$. Then $|A| \leq 3m_{(q-1)n/3}$.

Proof: WLOG $\gamma \neq 0$. Let $V \leq P_n^d$ denote the set of polynomials in P_n^d vanishing on the complement of $-\gamma A$. Let $P \in V$ have maximal support Σ . I claim that $|\Sigma| \geq \dim V$. Suppose otherwise and define the evaluation map $\Phi : V \rightarrow \mathbb{F}_q^\Sigma$ by $\Phi(Q) = (Q(x))_{x \in \Sigma}$ for all $Q \in V$. Then there exists $Q \neq 0$ such that Q vanishes (completely) on Σ . Pick $x_0 \in (-\gamma A) - \Sigma$ such that $Q(x_0) \neq 0$, so $P + Q$ has support containing $\Sigma \cup x_0$, absurd.

The complement of $-\gamma A$ has exactly $q^n - |A|$ elements, giving $q^n - |A|$ constraints. Thus, $\dim V$ is at least $m_d - (q^n - |A|) = m_d - q^n + |A|$. Combining with the previous result, we have $m_d - q^n + |A| \leq \dim V \leq |\Sigma|$. Let $\mathcal{S}(A) \subseteq \mathbb{F}_q^n$ denote the set of elements $\alpha a + \beta b$ for distinct $a, b \in A$. Suppose FTSOC $\mathcal{S}(A) \cap (-\gamma A) \neq \emptyset$, so $\alpha a + \beta b = -\gamma c$ for $a \neq b$, or $\alpha a + \beta b + \gamma c = 0$, absurd. Thus, P vanishes on $\mathcal{S}(A)$. By the previous result, $|\Sigma| \leq 2m_{d/2}$, so

$$\begin{aligned} |A| &\leq q^n - m_d + 2m_{d/2} \\ &\leq m_{n(q-1)-d} + 2m_{d/2} \end{aligned}$$

Take $d = \frac{2}{3}(q-1)n$, so

$$|A| \leq m_{n(q-1)-\frac{2}{3}(q-1)n} + 2m_{\frac{1}{3}(q-1)n} = 3m_{(q-1)n/3}.$$

2 Lower Bounds

Claim: Let $A \subseteq \mathbb{F}_3^n$ be a capset with size c^n . For every $\epsilon > 0$, there exists $M \geq 1$ such that for each $m \geq M$, there exists a capset A_m of size $(c - \epsilon)^m$ in $\mathbb{F}_3^m$.

Proof: Write $m = qn + r$, where $0 \leq r < n$. Let $A_m = A^q \times 0^r$, which is a capset by the previous claim. We compute:

$$|A_m|^{1/m} = c^{nq/m} = c^{nq/(nq+r)} > c^{nq/(nq+n)} = c^{q/(q+1)} = c \cdot c^{-1/(q+1)},$$

$\lim_{x \rightarrow \infty} c^{-1/(x+1)} = 1$, so pick M_0 such that $x \geq M_0 \rightarrow c^{-1/(x+1)} \geq 1 - \epsilon/c$. Let $M = nM_0$, so for $m \geq M$, we have $|A_m|^{1/m} > c - \epsilon$, as desired.

Claim: The bound from the product $A \times B$ is at most as good as the best of the bounds of A, B .

Proof: We wish to show $|A \times B|^{1/(n+m)} \leq \max\{|A|^{1/n}, |B|^{1/m}\}$. WLOG $|A|^{1/n} \leq |B|^{1/m}$. Then

$$\begin{aligned} |A \times B|^{1/(n+m)} &= |A|^{1/(n+m)} \cdot |B|^{1/(n+m)} = |B|^{\frac{n}{m} \cdot \frac{1}{n+m}} \cdot |B|^{\frac{1}{n+m}} \\ &= |B|^{\frac{n+m}{m} \cdot \frac{1}{n+m}} = |B|^{1/m}, \end{aligned}$$

as desired.

Extendable and Admissible sets. Cap sets $A_0, A_1, A_2 \subseteq \mathbb{F}_3^n$ are said to be extendable if the following conditions hold:

- (i). If $x, y \in A_0$ and $z \in A_1 \cup A_2$, then $x + y + z \neq 0$.
- (ii). If $x \in A_0, y \in A_1, z \in A_2$, then $x + y + z \neq 0$.

A subset $S \subseteq \{0, 1, 2\}^m$ is said to be admissible if the following conditions hold:

- (i). For all distinct $s, s' \in S$, there exist $i, j \in [m]$ such that $0 = s_i \neq s'_i$ and $s_j \neq s'_j = 0$.
- (ii). For all pairwise distinct $s, s', s'' \in S$, there exists $i \in [m]$ such that $\{s_i, s'_i, s''_i\} = \{0, 1, 2\}, \{0, 0, 1\}, \{0, 0, 2\}$, ie the indices are the same from the condition of extendable sets.

Claim: For a point $s \in \{0, 1, 2\}^m$, define $s(A_0, A_1, A_2) = \prod_{i=1}^m A_{s_i} \subseteq \mathbb{F}_3^{nm}$. Let $S(A_0, A_1, A_2) = \cup_{s \in S} s(A_0, A_1, A_2)$. If (A_0, A_1, A_2) is extendable and S is admissible, then $S(A_0, A_1, A_2)$ is a cap set.

Proof: Let $x, y, z \in S(A_0, A_1, A_2)$, not all the same such that $x + y + z = 0$. Let $x \in s(A_0, A_1, A_2)$ and define similarly t, u the corresponding vectors in $\{0, 1, 2\}^m$ associated to y, z . Fix $i \in [m]$. Let $I_i = \{s_i, t_i, u_i\}$. If $I_i = \{0, 1, 2\}$, we're done by condition (i) of Extendable sets. If $I_i = \{0, 0, 1\}, \{0, 0, 2\}$, then we're done by condition (ii) of extendable sets. Now we check when I_i contains two equal nonzero indices.

WLOG $1 = s_i = t_i \neq u_i$ (since 1, 2 have symmetric roles and x, y, z are symmetric). Thus, $s, t \neq u$. If s, t, u are distinct, by condition (ii) of Admissible sets and (i), (ii) of Extendable sets, we're done. Thus, assume $s = t$. Rename i to i_1 . By condition (i) of Admissible sets, pick i_2, i_3 such that $s_{i_2} = 0, u_{i_2} \neq 0, t_{i_3} = 0, u_{i_3} \neq 0$. Since $s = t$, we have $s_{i_2} = t_{i_2} = 0$, so we're done by condition (ii) of Extendable sets.

Finally, if I_i has one element (ie all of the indices are the same), then $x_i = y_i = z_i$ since each A_{s_i} is a cap set. Since this holds for all i , we have $x = y = z$, absurd.

Def: A recursively admissible set S is an admissible set with $|S| \geq 2$ such that one of the following properties hold for distinct $s, s' \in S$:

- (i). There exist $i, j \in [m]$ such that $\{s_i, s'_i\} = \{0, 1\}$ and $\{s_j, s'_j\} = \{0, 2\}$.
- (ii). There exists $k \in [m]$ such that $s_k = s'_k = 0$.

***Claim: If S is recursively admissible and (A_0, A_1, A_2) is extendable, then $(S(A_0, A_1, A_2), A_1^m, A_2^m)$ is extendable.

Proof: We show properties (i) and (ii) of extendable sets. Sps FTSOC $x, y \in S(A_0, A_1, A_2)$ and $z \in A_1^m$ such that $x + y + z = 0$ (the case $z \in A_2^m$ is analogous). Let $x \in s(A_0, A_1, A_2)$ and $y \in s'(A_0, A_1, A_2)$. Suppose $s \neq s'$. If (ii) of recursively admissible sets holds on s, s' , we're done by (i) of extendable sets. Suppose (i) of recursively admissible sets holds on s, s' and let i, j be such indices. At least one of the sets $\{s_i, s'_i, 1\}, \{s_j, s'_j, 1\}$ equals $\{0, 1, 2\}$, violating (ii) of extendable sets. Now suppose $s = s'$ and $i \in [m]$. If $s_i = s'_i = 0$, we're done by condition (i) of extendable sets. Thus, $s \in \{1, 2\}^m$, violating (i) of admissible sets (since $|S| \geq 2$).

Sps FTSOC $x \in s(A_0, A_1, A_2)$, $y \in A_1^m$, and $z \in A_2^m$ such that $x + y + z = 0$. If $s_i = 0$ for any $i \in [m]$, condition (ii) of extendable sets is violated. Thus, $s_i \in \{1, 2\}$ for all i , again a contradiction.

Claim: If we extend (A_0, A_1, A_2) by $S = I(m, w) \subseteq \{0, 1, 2\}^m$, where $|A_1| = |A_2|$, then $|S(A_0, A_1, A_2)| = \binom{m}{w} |A_0|^{m-w} |A_1|^w$.

Proof: Let $s, s' \in S$ be distinct. I claim that $s(A_0, A_1, A_2), s'(A_0, A_1, A_2)$ are disjoint. Pick i by (i) of admissible sets such that $0 = s_i \neq s'_i$. By a result of (i) for extendable sets, we have $A_{s_i} \cap A_{s'_i} = \emptyset$, so $s(A_0, A_1, A_2) \cap s'(A_0, A_1, A_2) = \emptyset$ (we intersect each multiplicand, of which index

i gives empty intersection). Then:

$$\begin{aligned} |S(A_0, A_1, A_2)| &= \sum_{s \in S} |s(A_0, A_1, A_2)| = \sum_{s \in S} |A_0|^{m-w} |A_1|^w \\ &= |S| \cdot |A_0|^{m-w} |A_1|^w = \binom{m}{w} |A_0|^{m-w} |A_1|^w \end{aligned}$$

***Claim (stronger statement): For all $m \geq 2$, there exists recursively admissible set $\tilde{I}(m, m-1)$.

Proof: We proceed by induction. For $m = 2$, we have $\tilde{I}(2, 1) = \{(1, 0), (0, 2)\}$. Suppose the existence of $\tilde{I}(m, m-1)$. Let S_{m+1} denote the set produced by adding 1 to the end of each vector of $\tilde{I}(m, m-1)$ and including a vector w with m twos followed by a zero. I claim that S_{m+1} is recursively admissible.

The supports of the elements are distinct, so condition (i) of admissible sets is satisfied. For a vector v , denote by $v \oplus a$ the vector v followed by $a \in \{0, 1, 2\}$. Let $v^i \in \tilde{I}(m, m-1)$ be the vector with i th component zero. Suppose $v^i \oplus 1, v^j \oplus 1, v^k \oplus 1 \in S_{m+1}$ are distinct. By the inductive hypothesis, we're done.

Suppose instead of $v^k \oplus 1$ we have w . Since $\tilde{I}(m, m-1)$ is recursively admissible and condition (ii) of recursively admissible sets can't be satisfied and the only component of v^i equal to 0 is at position i , we have $\{v_j^i, v_i^j\} = \{1, 2\}$. Thus, at least one of the sets $\{v_i^j, v_i^i, w_i\}, \{v_j^j, v_j^i, w_j\}$ equals $\{0, 1, 2\}$, satisfying condition (ii) of admissible sets for S_{m+1} .

Suppose $v^i \oplus 1, v^j \oplus 1 \in S_{m+1}$. By condition (i) of recursive admissible sets, the sets $\{v_i^i, v_i^j\}, \{v_j^i, v_j^j\}$ equal $\{0, 1\}, \{0, 2\}$ in some order. Suppose $v^i \oplus 1, u \in S_{m+1}$. Then $\{v_i^i, w_i\}, \{v_{m+1}^i, w_{m+1}\}$ equal $\{0, 2\}, \{1, 0\}$, so condition (i) of recursive admissible sets is satisfied on S_{m+1} , completing the induction.

Claim: If S, T are admissible, so is their direct product $S \times T$.

Proof: Let $S \subseteq \{0, 1, 2\}^m$ and $T \subseteq \{0, 1, 2\}^{m'}$. Suppose $(s, t), (s', t') \in S \times T$ are distinct. If $s \neq s'$, pick i, j such that $0 = s_i \neq s'_i$ and $s_j \neq s'_j = 0$. Since we have $(s, t)_i = s_i$ and similar formulas, we're done. If $t \neq t'$, apply formulas of the type $(s, t)_{i+m} = t_i$, so we're also done.

Suppose $(s, t), (s', t'), (s'', t'') \in S \times T$ are distinct. If s, s', s'' or t, t', t'' are distinct, we're done. If any of s, s', s'' and t, t', t'' are all the same, then the other set of indices are distinct, so we're done. It remains to check when $\{s, s', s''\}, \{t, t', t''\}$ both have two elements. WLOG $s = s' \neq s''$, so pick i such that $0 = s_i \neq s''_i$. Then $\{(s, t)_i, (s', t')_i, (s'', t'')_i\}$ equals one of $\{0, 0, 1\}, \{0, 0, 2\}$.

Claim: If $S_0, S_1, S_2 \subseteq \{0, 1, 2\}^m$ is a meta-extendable collection of admissible sets and $T \subseteq \{0, 1, 2\}^r$ is meta-admissible, then $T(S_0, S_1, S_2)$ is admissible.

Proof: Suppose we have distinct $x \in t(S_0, S_1, S_2), y \in t'(S_0, S_1, S_2)$. If $t = t'$, pick i such that $x_i \neq y_i$. Then there exists j such that $0 = (x_i)_j \neq (y_i)_j$ and $(x_i)_k \neq (y_i)_k = 0$.

3 Lower Bound Computations

Suppose we have cap sets $A_0, A_1, A_2, A'_1, A'_2$ such that $|A_1| = |A_2|$ and $|A_1| + |A_2| = |A'_1| + |A'_2|$, and $S = I(m, w)$ is admissible. What can we say about $|S(A_0, A_1, A_2)|, |S(A_0, A'_1, A'_2)|$?

For a vector s , denote by $w_0(s), w_1(s), w_2(s)$ the count of components equal to 0, 1, 2, resp. We have

$$|S(A_0, A_1, A_2)| = \sum_{s \in S} |s(A_0, A_1, A_2)| = \sum_{s \in S} |A_0|^{m-w} |A_1|^{w_1(s)} |A_2|^{w_2(s)}$$

Suppose S is symmetric, so for $s \in S$, there exists $s' \in S$ with $w_1(s') = w_2(s)$ and $w_2(s') = w_1(s)$. Fix values $w_1(s) \leq w_2(s)$, let $d = w_2(s) - w_1(s)$, and consider the balance:

$$\begin{aligned} |A'_1|^{w_1(s)} |A'_2|^{w_2(s)} + |A'_1|^{w_2(s)} |A'_2|^{w_1(s)} &? |A_1|^{w_1(s)} |A_2|^{w_2(s)} + |A_1|^{w_2(s)} |A_2|^{w_1(s)} \\ |A'_1|^{w_1(s)} |A'_2|^{w_1(s)} (|A'_2|^d + |A'_1|^d) &? |A_1|^{w_1(s)} |A_2|^{w_1(s)} (|A_2|^d + |A_1|^d) \end{aligned}$$

The balancing shifts larger right with $|A'_1|^{w_1(s)} |A'_2|^{w_1(s)} ? |A_1|^{w_1(s)} |A_2|^{w_1(s)}$ and larger left with $|A'_2|^d + |A'_1|^d ? |A_2|^d + |A_1|^d$, so we wish to maximize the difference between $w_1(s), w_2(s)$.